GLS UNIVERSITY
MASTER OF COMPUTER APPLICATIONS (MCA) SEMESTER I
CAPSTONE PROJECT – I PROJECT PROPOSAL

WELLNEXA AI

An AI-Powered Integrated Personal Healthcare & Wellness Management Platform

"One Platform. Your Complete Health Journey."

Group Project – 5 Members | Academic Year: 2026–27

1. Project Overview

WellNexa AI is an all-in-one personal healthcare and wellness management platform. It combines medical records, medical-report understanding, medicines, nutrition, fitness, mental wellness, preventive health tracking, appointments, reminders, and AI-powered health education within one connected system. A single personal health profile allows authorized information to be organized and connected across modules.

2. Abstract

WellNexa AI is a proposed AI-powered preventive healthcare and wellness platform designed to address fragmented personal health information and the difficulty users face in understanding and managing everyday health needs. It provides a centralized health profile, medical-report analysis, prescription and medicine management, nutrition planning, fitness and lifestyle tracking, mental-wellness tools, preventive analytics, appointments, reminders, and controlled healthcare-professional interaction. OCR, NLP, LLMs, RAG, recommendation systems, predictive analytics, and optional computer vision will be explored. The platform is intended for health management and education, not diagnosis or treatment, with strong emphasis on privacy, security, transparency, and responsible AI.

3. Real-World Problem

People commonly manage health using disconnected tools. Medical reports may be physical documents or PDFs, prescriptions can be difficult to understand, medicines and appointments may be forgotten, and nutrition, exercise, sleep, and mental wellness are often tracked separately. Users may also struggle to compare health information over time. Existing applications frequently focus on one area. WellNexa AI

addresses this fragmentation by integrating these needs through one platform and a common health profile.

4. Proposed Solution

WellNexa AI will create a centralized personal health ecosystem. Users maintain one authorized health profile. The platform connects report understanding, health tracking, medicine management, nutrition, wellness, reminders, appointments, and AI assistance. For example, a user can upload a laboratory report, use OCR to extract information, receive an AI-assisted explanation, store the data in a health timeline, compare it with previous results, and selectively share information with a healthcare professional.

5. Project Objectives

- Address a genuine real-world problem in personal health information management.
- Create a centralized platform for health records, wellness activities, and healthcare tasks.
- Use OCR and AI to extract and explain supported medical reports and prescriptions.
- Provide personalized nutrition, lifestyle, reminder, and preventive-health assistance.
- Track selected health metrics and visualize trends.
- Develop an LLM/RAG-based health-education assistant.
- Enable controlled sharing of authorized information with healthcare professionals.
- Demonstrate AI, software engineering, database, security, testing, and teamwork skills.

6. Scope of the Project

- Personal health profile and history.
- Medical document upload, OCR, and AI-assisted explanation.
- Prescription and medicine management.
- Medicine, refill, appointment, checkup, and wellness reminders.
- Nutrition logging, meal planning, and dietary suggestions.
- Fitness and lifestyle tracking.
- Mental-wellness journaling and mood/stress tracking.
- Preventive-health analytics and trend monitoring.
- Doctor/healthcare-professional portal with controlled sharing.
- AI health assistant using LLM and RAG.
- Secure authentication, authorization, data management, and audit logging.

7. Major Modules and Features

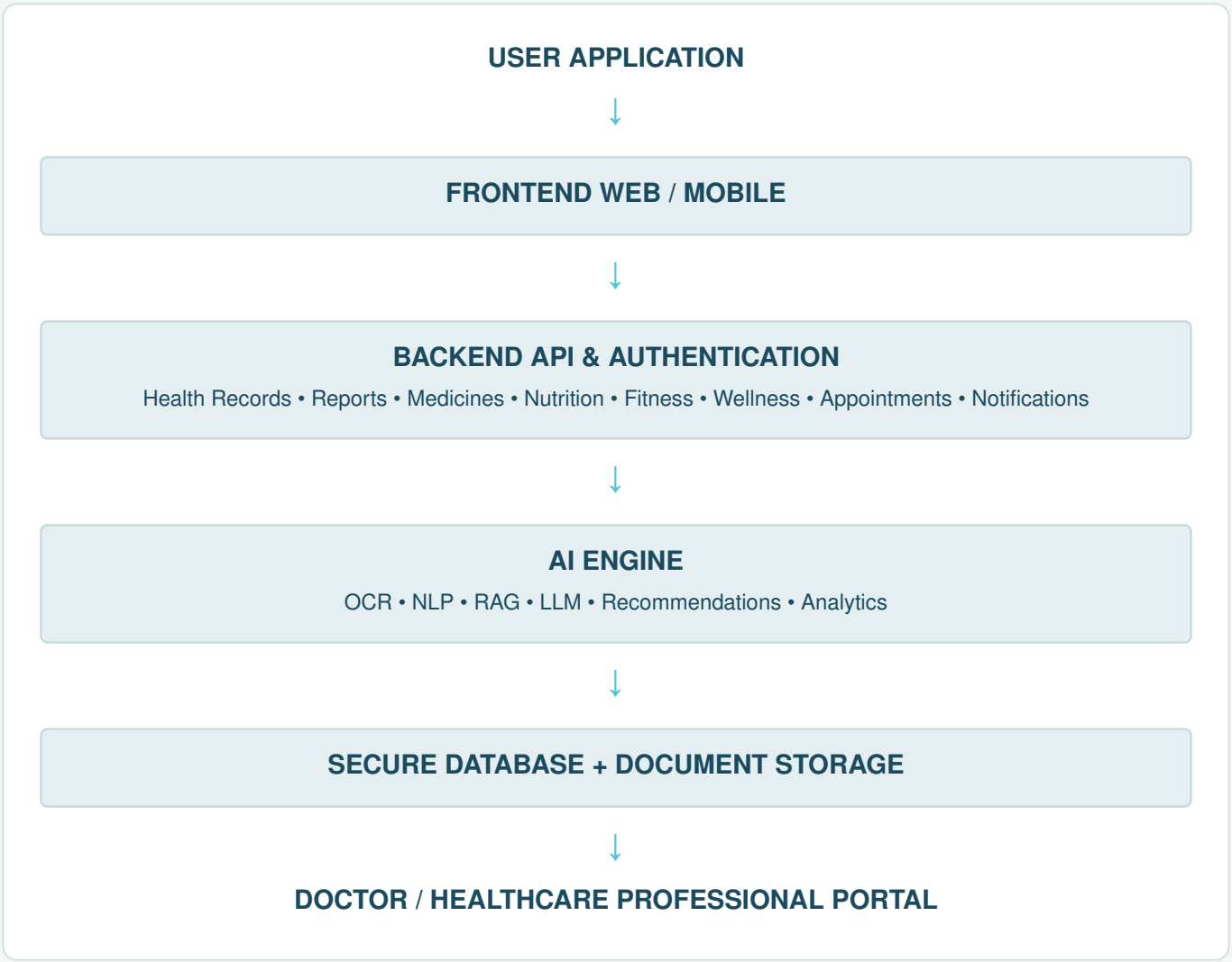
Module	Main Features
Personal Health Profile	Age, height, weight, BMI, BP, glucose, allergies, lifestyle, goals, authorized history.
AI Medical Report Analyzer	OCR, report explanation, reference-range highlighting, comparison and trends.
Prescription & Medicine Manager	Prescription OCR, medicine schedules, reminders, refill alerts, medication history.
AI Nutrition Planner	Food logging, calorie/nutrient tracking, meal suggestions, grocery planning, optional food recognition.
Fitness & Lifestyle Tracker	Steps, exercise, sleep, water, weight, activity goals and progress.
Mental Wellness	Mood journal, stress tracking, reflective prompts, relaxation exercises and wellness trends.
Doctor Portal	Controlled record sharing, patient summaries, appointments and professional review.
AI Health Assistant	General health education, document explanation, authorized-record questions, RAG/LLM assistance.
Smart Reminders	Medicine, appointments, checkups, lab tests, vaccinations, refills and wellness reminders.
Health Dashboard	Unified medical, nutrition, lifestyle, wellness, medication and appointment overview.

8. AI Technologies

Technology	Application
OCR	Extract text and structured information from reports and prescriptions.
NLP	Process and simplify health-related text and terminology.
LLM	Power the conversational WellNexa AI health assistant.
RAG	Ground responses in approved health sources and authorized user documents.
Recommendation System	Personalized nutrition, lifestyle, reminder and wellness suggestions.
Predictive Analytics	Health trend analysis and appropriate validated risk calculations.
Computer Vision (Optional)	Food recognition for nutrition logging.

9. System Architecture

The proposed architecture uses a modular client-server design. The frontend communicates with backend APIs. Backend services manage authentication, authorization, business logic, records, documents, notifications, and integrations. An AI layer manages OCR, LLM/RAG, recommendations, and analytics. Health data and documents are stored securely with access controls and audit mechanisms.


USER APPLICATION
FRONTEND WEB / MOBILE
BACKEND API & AUTHENTICATION
Health Records • Reports • Medicines • Nutrition • Fitness • Wellness • Appointments • Notifications
AI ENGINE
OCR • NLP • RAG • LLM • Recommendations • Analytics
SECURE DATABASE + DOCUMENT STORAGE
DOCTOR / HEALTHCARE PROFESSIONAL PORTAL

10. Suggested Technology Stack

Layer	Suggested Technology
Frontend	React / Next.js or Flutter
Backend	Python FastAPI or Node.js + Express
Database	PostgreSQL or MongoDB
AI/ML	Python, scikit-learn, suitable LLM API or local model
RAG	FAISS, Chroma, or pgvector
OCR	Tesseract or suitable OCR service
Authentication	JWT/OAuth with role-based access control
Visualization	Chart.js / Recharts
Deployment	Suitable cloud platform
Version Control	Git and GitHub/GitLab
Testing	Postman, unit, integration and UI testing

11. Functional Requirements

1. User registration, authentication and profile management.
2. Management of authorized health information.
3. Upload of supported medical reports and prescriptions.
4. OCR extraction from supported documents.
5. AI-assisted report summaries and explanations.
6. Medicine schedules and reminders.
7. Nutrition, fitness, lifestyle and wellness tracking.
8. Health metric visualization and historical trends.
9. Appointment management and controlled professional interaction.
10. AI health-education assistant.
11. Notifications and appropriate audit logs.

12. Non-Functional Requirements

- **Security:** authentication, authorization, encryption and access controls.
- **Privacy:** consent, data minimization and sharing controls.
- **Usability:** simple, accessible and understandable interfaces.
- **Performance:** efficient responses for expected project-scale usage.
- **Reliability:** consistent records, reminders and transactions.
- **Scalability:** support future wearable, telemedicine and AI modules.
- **Maintainability:** modular, documented, tested and version-controlled code.
- **Transparency:** clearly label AI-generated information and limitations.

13. Five-Member Team Structure

Member	Role	Responsibilities
Member 1	Frontend & UI/UX	Web/mobile interface, dashboards, forms and usability.
Member 2	Backend & APIs	Authentication, APIs, business logic, notifications and integrations.
Member 3	Database & Security	Schema, health records, access control, security and audit logs.
Member 4	AI/ML & OCR	OCR, LLM, RAG, recommendations and AI evaluation.
Member 5	Testing, Analytics & Deployment	Testing, analytics, deployment, QA, documentation and coordination.

14. Capstone Project–I Deliverables

- Problem identification and background study.
- Literature survey and existing-system analysis.
- Objectives, scope, assumptions and constraints.
- Software Requirement Specification (SRS).
- Use-case and requirement models.
- System architecture and module design.
- ER diagram and database schema.

- Data-flow and other design diagrams.
- UI/UX wireframes and prototype.
- AI/OCR feasibility study and initial proof of concept.
- Project timeline, task allocation, testing strategy and documentation.

15. Future Scope

- Smartwatch and wearable-device integration.
- Voice-based AI health assistant.
- Multiple Indian and international languages.
- Telemedicine and secure online consultations.
- Hospital/clinic integrations where appropriate.
- Pharmacy and medicine-delivery integrations.
- Health-insurance integrations.
- Family/caregiver access with explicit consent.
- IoT health monitoring.
- Advanced personalized analytics and recommendation models.

16. Safety, Privacy & Ethical Considerations

WellNexa AI will be positioned as a preventive healthcare and health-management platform, not a substitute for a qualified healthcare professional. It should not independently diagnose diseases or prescribe treatment. AI outputs should be informational, transparent about limitations, and appropriately refer users to professional care for clinical decisions.

- Consent-based access and sharing.
- Authentication, authorization, encryption and secure storage.
- Audit logs for sensitive actions.
- Data minimization.
- Clear labeling of AI-generated content.
- Evaluation of hallucination, bias and reliability risks.
- Trusted sources for RAG-based health information.
- Appropriate safety guidance for situations requiring professional attention.

17. What Makes WellNexa AI Unique

WellNexa AI is differentiated by integration. Medical reports, medicines, nutrition, fitness, mental wellness, preventive health, reminders, appointments and AI assistance are connected through one personal health profile.

1. Upload a medical report.
2. OCR extracts relevant information.
3. AI explains the report in understandable language.
4. Information is added to the authorized health timeline.
5. Previous results can be compared and trends visualized.
6. Relevant nutrition/lifestyle suggestions can be generated where appropriate.
7. Follow-up reminders can be created.
8. Selected information can be shared with a healthcare professional.

This connected workflow makes WellNexa AI more than a medicine reminder, fitness tracker, medical chatbot, or report reader; it is designed as an integrated personal health ecosystem.

18. Conclusion

WellNexa AI is a comprehensive MCA Capstone Project addressing a genuine challenge in personal health management. By bringing medical records, AI-assisted report understanding, medicines, nutrition, fitness, mental wellness, preventive insights, appointments, reminders, and professional interaction into one platform, it offers substantial technical scope and practical value.

Semester I will establish the problem definition, requirements, SRS, architecture, database design, UI/UX prototype, AI feasibility and initial proof of concept. Future phases can expand the platform with wearables, voice interaction, telemedicine, advanced analytics and additional integrations.

Final Project Identity

Project Name: WellNexa AI

Full Title: An AI-Powered Integrated Personal Healthcare & Wellness Management Platform

Tagline: *One Platform. Your Complete Health Journey.*